Human Activity Recognition using Wi-Fi CSI

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Motivation

Human Activity Recognition (HAR) has several applications such as "patient fall detection", "surveillance", etc..

Types of HAR systems:

1. Device-assisted HAR system
2. Device-free HAR system

Wi-Fi CSI based HAR systems are recently introduced as a device-free HAR system to detect human's presence and activity recognition. Due to ubiquity of Wi-Fi networks, such systems are expected to be widely adopted.

Extended Applications: human identification [1], fire detection [2]

Introduction

- Movement of human body causes a change in the wireless signal reflections, which results in variations in the CSI
- By analyzing the CSI in presence of certain activities (e.g. standing, sitting, walking, falling) and comparing them against stored models, human activity can be recognized.
- Deep-learning (DL) techniques are employed to build robust models for device-free prediction of human activity.

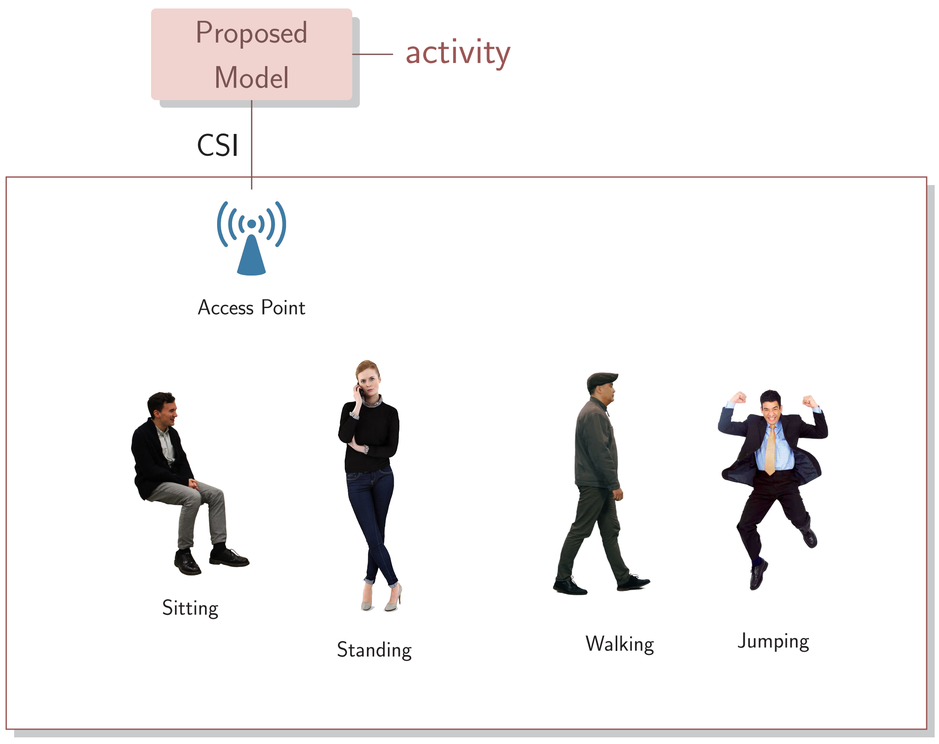


Figure: Human Activity Recognition

Proposed Scheme

- **Data Acquisition**
  - CSI data acquired through Intel 5300 NIC with Linux 802.11n CSI tool, constitute amplitude and phase of the received signals. Datasets are provided by authors in [3].
- **Preprocessing**
  - Handling of missing data
- **Feature Extraction**
  - Dimensionality reduction
- **Training and Validation**
  - Long Short Term Recurrent Neural Networks (LSTM-RNN) are used.
  - Training data constitute input data points with associated output values.
  - A single input data point is a 90 dimensional vector which is actually the signal amplitude data acquired using a Wi-Fi device with 3 antennas.
  - Each input data point is associated with one of the seven (7) output labels (no activity, lie down, fall, walk, run, sit down, stand up).

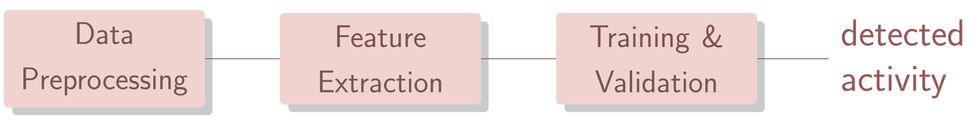


Figure: Proposed HAR Scheme

- **Tensorflow (Google) is used to implement the proposed model**

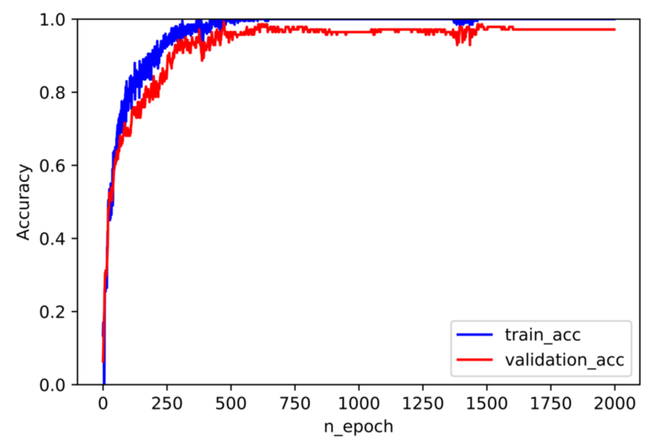
Results: Confusion Matrix

- Confusion Matrix provides information about the correct and wrong predictions.

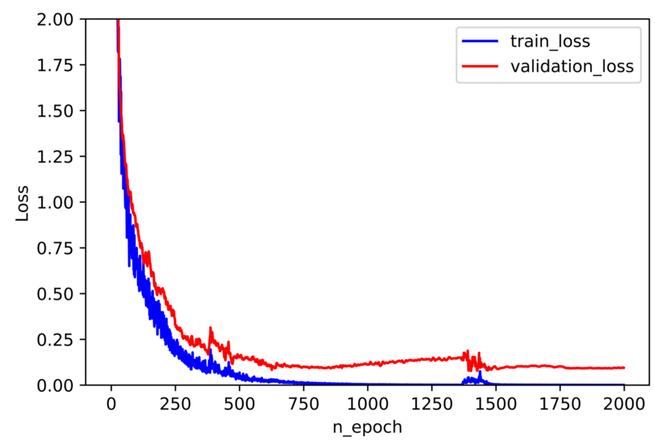
	Lie down	Fall	Walk	Run	Sit down	Stand up
Lie down	150	0	3	2	4	0
Fall	0	85	7	0	0	0
Walk	0	1	432	1	9	0
Run	4	1	0	95	15	2
Sit down	3	1	5	9	416	1
Stand up	1	0	2	0	4	94

Table: Confusion Matrix

Results: Model Performance



(a) Accuracy



(b) Loss

Conclusions and Future Work

- The accuracy of the the model was 93% over validation data.
- The training time for the model is around 6-8 hours on PC with Intel Core i7-4770 3.4GHz processor, 8GB RAM.
- Model used is similar to one presented in [3], with minor changes.
- Ongoing work.. To design a faster and robust model using least possible amount of data without compromising accuracy

References

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